Astronomy and Astrophysics Olympiad (Observation)

A Comprehensive Note Created by Former Hong Kong Team Members of IOAA

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1 Sky Map

1.1 Introduction

A sky map (or star chart) is a graphical representation of the night sky as seen from a specific location on Earth at a given time.

1.2 Equatorial Coordinates

Sky maps are typically based on a coordinate system known as equatorial coordinates, which is analogous to geographic latitude and longitude. This system uses two main parameters:

- Right Ascension (RA) is similar to longitude on Earth. It measures the angular distance of an object eastward along the celestial equator. RA is usually expressed in hours, minutes, and seconds, where

$$1 \text{ hour} = 15^\circ$$

- Declination (Dec) is similar to latitude on Earth. It measures the angular distance of an object north or south of the celestial equator. Declination is expressed in degrees, where:

$$\text{North of the celestial equator} = +\text{degrees}, \quad \text{South of the celestial equator} = -\text{degrees}$$

1.3 Celestial Equator

The celestial equator is the projection of Earth's equator onto the celestial sphere. It divides the sky into the northern and southern hemispheres and runs directly through the center of the map.

1.4 Ecliptic Plane

The ecliptic is the apparent path that the Sun traces across the sky over the course of a year. It is inclined at an angle of approximately 23.5° to the celestial equator. The ecliptic plane is often shown as a dashed or thick line on sky maps and is crucial for locating the positions of the planets.

2 Constellations

Constellations are groupings of stars that are often named after mythological figures, animals, or objects. These star patterns have been used by various civilizations for navigation, timekeeping, and storytelling. Today, the International Astronomical Union (IAU) recognizes 88 official constellations.

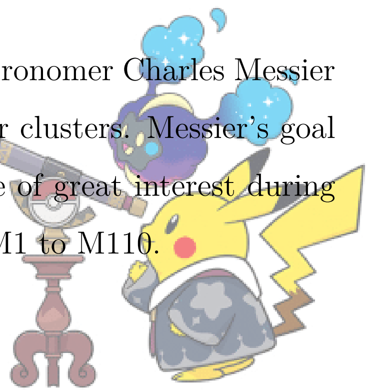
Constellation	**Abbreviation**
Andromeda	And
Antlia	Ant
Apus	Apus
Aquarius	Aqr
Aquila	Aql
Aries	Ari
Auriga	Aur
Boötes	Boo
Caelum	Cae
Camelopardalis	Cam
Cancer	Cnc
Canes Venatici	CVn
Canis Major	CMa
Canis Minor	CMi
Capella	Cap
Carina	Car
Cassiopeia	Cas
Centaurus	Cen
Cepheus	Cep
Cetus	Cet
Chamaeleon	Cha
Circinus	Cir
Columba	Col
Coma Berenices	Com
Corona Australis	CrA
Corona Borealis	CrB

Constellation	**Abbreviation**
Corvus	Crv
Crater	Crt
Cygnus	Cyg
Delphinus	Del
Dorado	Dor
Draco	Dra
Equuleus	Equ
Eridanus	Eri
Fornax	For
Gemini	Gem
Grus	Gru
Hercules	Her
Horologium	Hor
Hydra	Hya
Hydrus	Hyi
Indus	Ind
Lacerta	Lac
Leo	Leo
Leo Minor	LMi
Lupus	Lup
Lynx	Lyn
Lyra	Lyr
Mensa	Men
Microscopium	Mic
Monoceros	Mon
Musca	Mus
Norma	Nor
Octans	Oct
Ophiuchus	Oph
Orion	Ori
Pavo	Pav

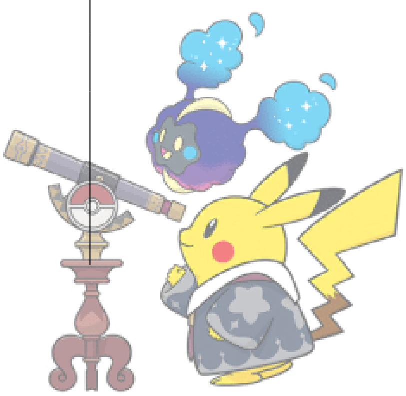
Constellation	**Abbreviation**
Pegasus	Peg
Perseus	Per
Phoenix	Phe
Pictor	Pic
Pisces	Pis
Piscis Austrinus	PsA
Portion of the Southern Cross	Crux
Sagitta	Sge
Sagittarius	Sgr
Scorpius	Sco
Sculptor	Scl
Scutum	Scu
Serpens	Ser
Sextans	Sex
Taurus	Tau
Telescopium	Tel
Triangulum	Tri
Triangulum Australe	TrA
Tucana	Tuc
Ursa Major	UMa
Ursa Minor	UMi
Vela	Vel
Virgo	Vir
Volans	Vol

3 Messier Objects

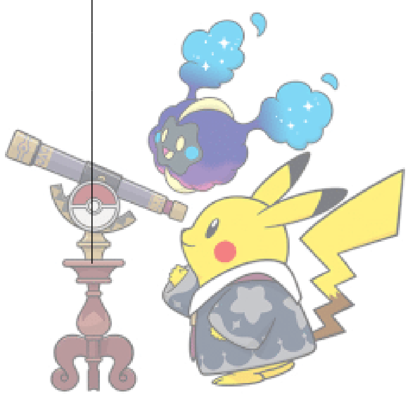
The Messier Objects are a list of astronomical objects cataloged by the French astronomer Charles Messier in the 18th century. The catalog consists primarily of galaxies, nebulae, and star clusters. Messier's goal was to compile a list of objects that might be confused with comets, which were of great interest during his time. There are 110 Messier Objects in total, each assigned a number from M1 to M110.



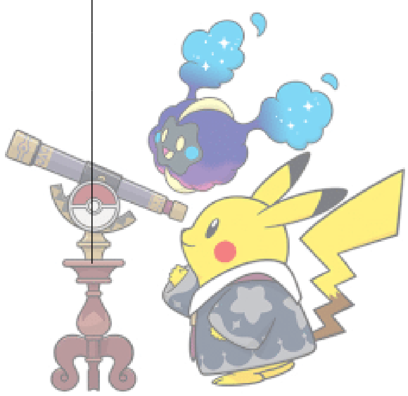
M Number	Object Name
1	M1: Crab Nebula
2	M2: Globular Cluster in Aquarius
3	M3: Globular Cluster in Canes Venatici
4	M4: Globular Cluster in Scorpius
5	M5: Globular Cluster in Serpens
6	M6: Butterfly Cluster
7	M7: Ptolemy's Cluster
8	M8: Lagoon Nebula
9	M9: Globular Cluster in Ophiuchus
10	M10: Globular Cluster in Ophiuchus
11	M11: Wild Duck Cluster
12	M12: Globular Cluster in Ophiuchus
13	M13: Hercules Cluster
14	M14: Globular Cluster in Ophiuchus
15	M15: Globular Cluster in Pegasus
16	M16: Eagle Nebula
17	M17: Omega Nebula
18	M18: Open Cluster in Sagittarius
19	M19: Globular Cluster in Ophiuchus
20	M20: Trifid Nebula
21	M21: Open Cluster in Sagittarius
22	M22: Globular Cluster in Sagittarius
23	M23: Open Cluster in Sagittarius
24	M24: Small Sagittarius Star Cloud
25	M25: Open Cluster in Sagittarius
26	M26: Open Cluster in Scutum
27	M27: Dumbbell Nebula
28	M28: Globular Cluster in Sagittarius
29	M29: Open Cluster in Cygnus
30	M30: Globular Cluster in Capricornus
31	M31: Andromeda Galaxy



M Number	Object Name
32	M32: Elliptical Galaxy in Andromeda
33	M33: Triangulum Galaxy
34	M34: Open Cluster in Perseus
35	M35: Open Cluster in Gemini
36	M36: Open Cluster in Auriga
37	M37: Open Cluster in Auriga
38	M38: Open Cluster in Auriga
39	M39: Open Cluster in Cygnus
40	M40: Double Star (Winnecke 4)
41	M41: Open Cluster in Canis Major
42	M42: Orion Nebula
43	M43: Nebula in Orion
44	M44: Beehive Cluster
45	M45: Pleiades
46	M46: Open Cluster in Puppis
47	M47: Open Cluster in Puppis
48	M48: Open Cluster in Hydra
49	M49: Elliptical Galaxy in Virgo
50	M50: Open Cluster in Monoceros
51	M51: Whirlpool Galaxy
52	M52: Open Cluster in Cassiopeia
53	M53: Globular Cluster in Coma Berenices
54	M54: Globular Cluster in Sagittarius
55	M55: Globular Cluster in Sagittarius
56	M56: Globular Cluster in Lyra
57	M57: Ring Nebula
58	M58: Barred Spiral Galaxy in Virgo
59	M59: Elliptical Galaxy in Virgo
60	M60: Elliptical Galaxy in Virgo
61	M61: Spiral Galaxy in Virgo
62	M62: Globular Cluster in Ophiuchus



M Number	Object Name
63	M63: Sunflower Galaxy
64	M64: Black Eye Galaxy
65	M65: Spiral Galaxy in Leo
66	M66: Spiral Galaxy in Leo
67	M67: Open Cluster in Cancer
68	M68: Globular Cluster in Hydra
69	M69: Globular Cluster in Sagittarius
70	M70: Globular Cluster in Sagittarius
71	M71: Globular Cluster in Sagitta
72	M72: Globular Cluster in Aquarius
73	M73: Group of Stars in Aquarius
74	M74: Spiral Galaxy in Pisces
75	M75: Globular Cluster in Sagittarius
76	M76: Little Dumbbell Nebula
77	M77: Seyfert Galaxy in Cetus
78	M78: Reflection Nebula in Orion
79	M79: Globular Cluster in Lepus
80	M80: Globular Cluster in Scorpius
81	M81: Bode's Galaxy
82	M82: Cigar Galaxy
83	M83: Southern Pinwheel Galaxy
84	M84: Elliptical Galaxy in Virgo
85	M85: Elliptical Galaxy in Coma Berenices
86	M86: Elliptical Galaxy in Virgo
87	M87: Giant Elliptical Galaxy in Virgo
88	M88: Spiral Galaxy in Virgo
89	M89: Elliptical Galaxy in Virgo
90	M90: Spiral Galaxy in Virgo
91	M91: Spiral Galaxy in Leo
92	M92: Globular Cluster in Hercules
93	M93: Open Cluster in Puppis



M Number	Object Name
94	M94: Spiral Galaxy in Canes Venatici
95	M95: Spiral Galaxy in Leo
96	M96: Spiral Galaxy in Leo
97	M97: Owl Nebula
98	M98: Spiral Galaxy in Virgo
99	M99: Spiral Galaxy in Coma Berenices
100	M100: Spiral Galaxy in Coma Berenices
101	M101: Pinwheel Galaxy
102	M102: Galaxy in Draco
103	M103: Open Cluster in Cassiopeia
104	M104: Sombrero Galaxy
105	M105: Elliptical Galaxy in Leo
106	M106: Spiral Galaxy in Canes Venatici
107	M107: Globular Cluster in Ophiuchus
108	M108: Spiral Galaxy in Ursa Major
109	M109: Barred Spiral Galaxy in Ursa Major
110	M110: Dwarf Elliptical Galaxy in Andromeda

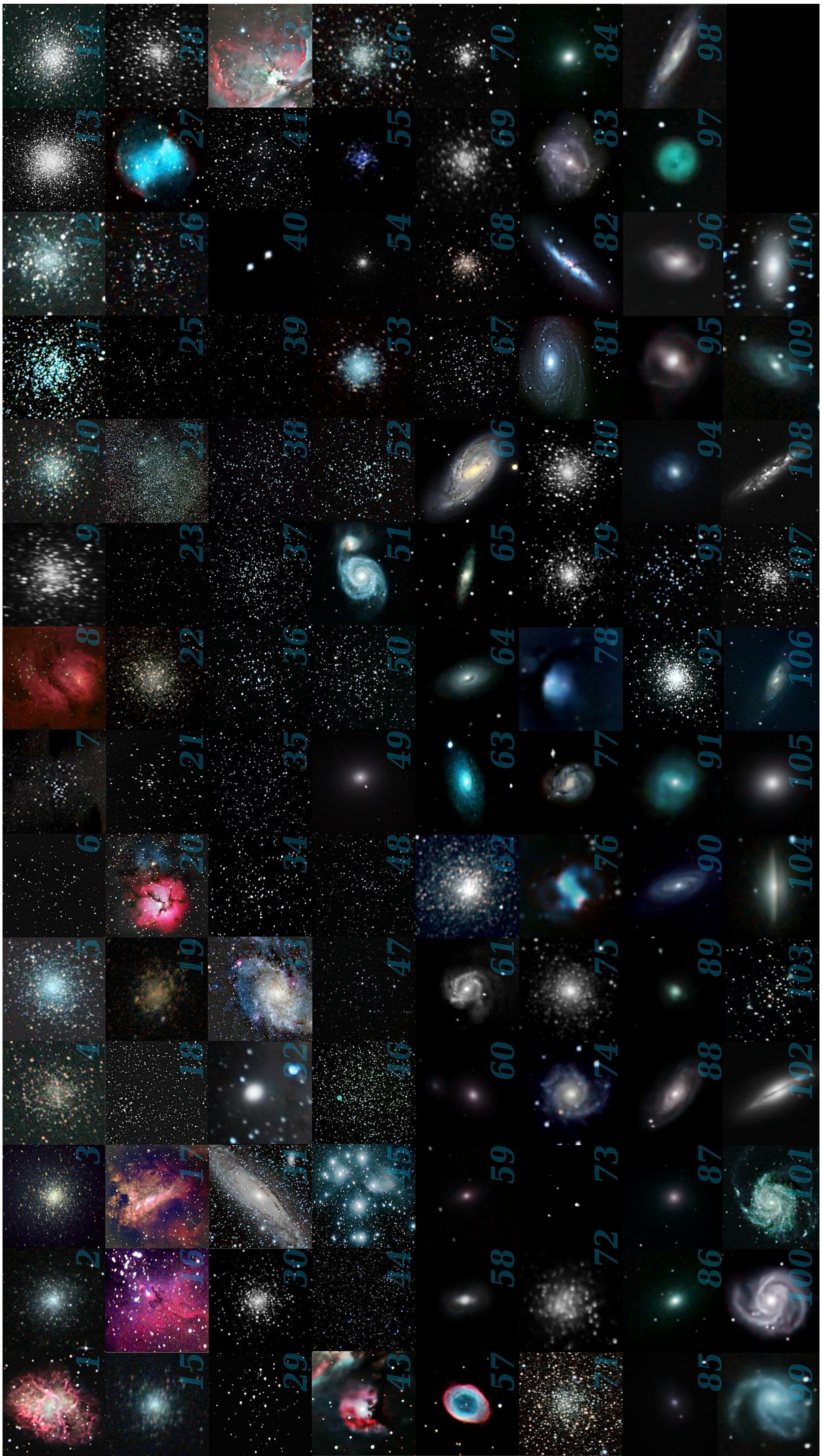


Figure 1: Photo of Each Messier Object from Wikipedia